

Tensor Algebra for the Encryption Algorithm: Extension to Image, Audio, Text, and Other Modalities

Alshicrypt Multimodal

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Objects

- V : tensor space for a modality
- $x \in V$: data tensor
- $T : V \rightarrow V$: tensor morphism / endomorphism
- $E : V \rightarrow V$: learned Encrypter
- $D : V \rightarrow V$: learned Decrypter

Tensor-Morphism Picture

$$\begin{array}{ccc} x & \xrightarrow{T} & T(x) \\ E \downarrow & \nearrow \approx & \\ E(x) & & \end{array}$$

This expresses the intended learned relationship

$$E(x) \approx T(x).$$

Learned Reconstruction Picture

$$x \xrightarrow{E} E(x) \xrightarrow{D} D(E(x))$$

with the intended reconstruction relation

$$D(E(x)) \approx x.$$

Interpretation

The object x is a full tensor, and the morphism T acts on that tensor as a whole. This is the tensor-algebra viewpoint: we treat the modality object itself as a tensor in a tensor space V and study endomorphisms

$$T : V \rightarrow V.$$

In the learned system, the Encrypter E is trained to approximate the forward tensor morphism, while the Decrypter D is trained to map the transformed tensor back toward the original.

Multimodal Examples

- For images, x may be a tensor with RGB values in $\{0, \dots, 255\}^{H \times W \times 3}$ together with a preserved alpha mask.
- For audio, x may be a waveform tensor or feature tensor in an analogous tensor space.

The current repository demonstrates the image case first, but the same tensor-space framing is intended to extend to audio next and, more generally, to text and other modalities that admit a tensor representation.